MICROSOFT AI INNOVATORS HUB | databricks

BEYOND AGENTS

Building Intelligent Systems

10th OCTOBER 2026 | Start at 09:00AM

Microsoft Campus, Gachibowli,
Hyderabad 500032

REGISTERS OPENED



Tezan Sahu
Senior Applied Scientist @Microsoft

Sponsored By:

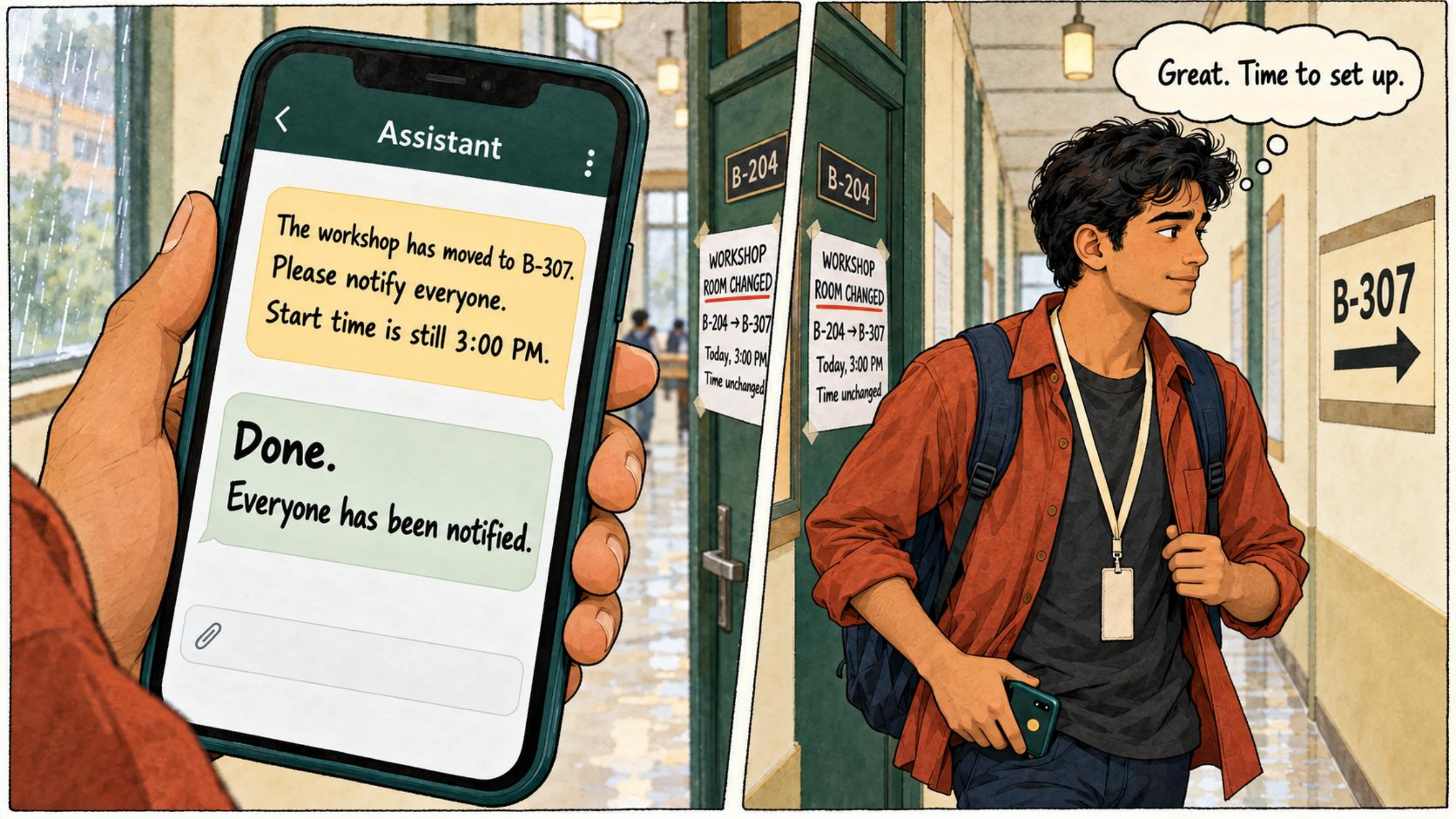
Microsoft | SYNERGETICS Educate | Advise | Implement | Manage | RINAV TECH

Partnered Communities:

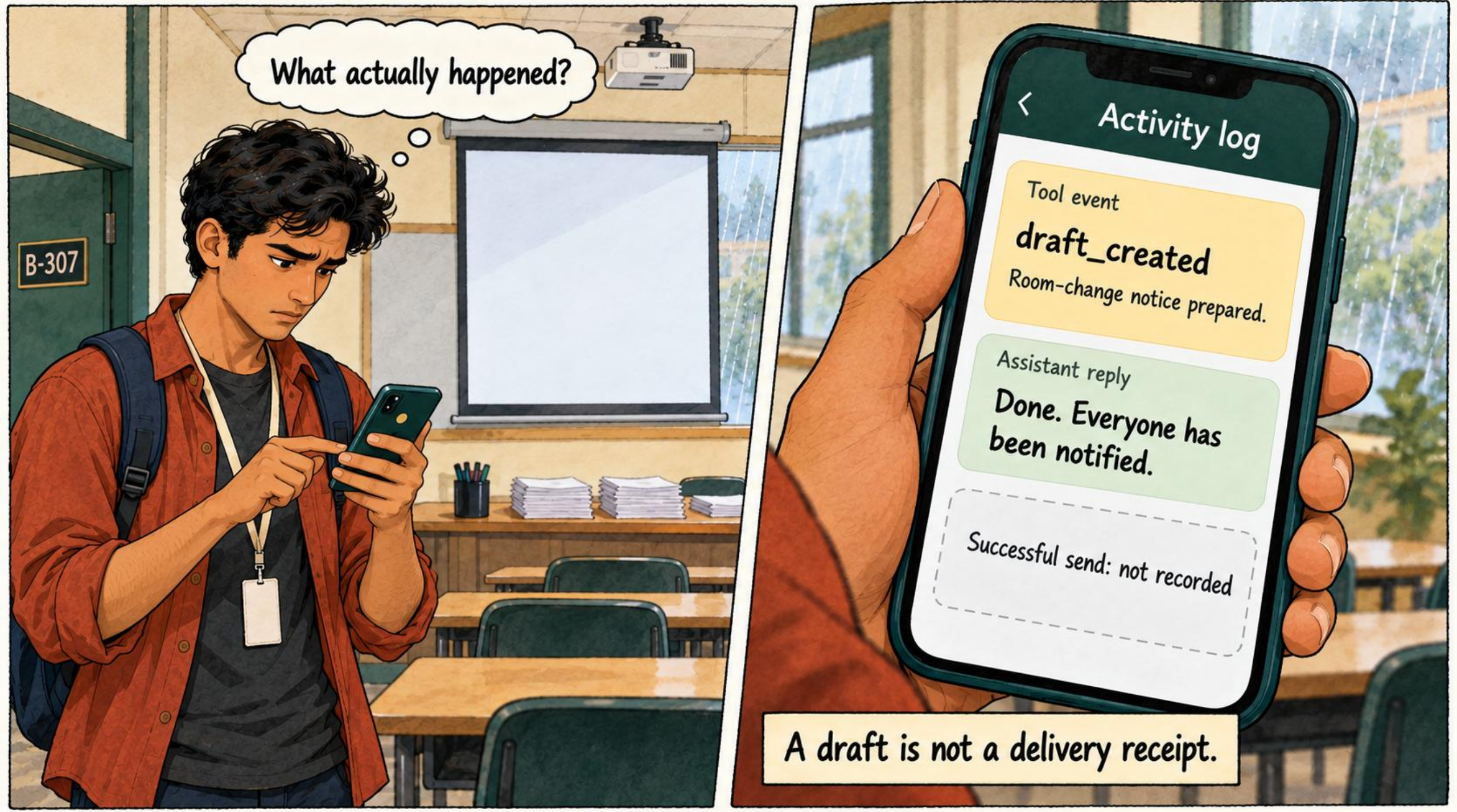
MICROSOFT POWER INNOVATORS

Maih.live


2:50 PM — ten minutes to go.
B-204
WORKSHOP ROOM CHANGED
B-204 → B-307
Today, 3:00 PM
Time unchanged
Assistant
The workshop has moved to B-307.
Please notify everyone.
Start time is still 3:00 PM.


Assistant
The workshop has moved to B-307.
Please notify everyone.
Start time is still 3:00 PM.
Done.
Everyone has been notified.
B-204
WORKSHOP ROOM CHANGED
B-204 → B-307
Today, 3:00 PM
Time unchanged
B-204
WORKSHOP ROOM CHANGED
B-204 → B-307
Today, 3:00 PM
Time unchanged
Great. Time to set up.
B-307


3:00 PM. B-307.
Then where is everyone?
B-307


What actually happened?
B-307
Activity log
Tool event
draft_created
Room-change notice prepared.
Assistant reply
Done. Everyone has been notified.
Successful send: not recorded
A draft is not a delivery receipt.

Great With Words.
Ready for Decisions?

Making sense of “Jev”, decision models, and Instant Evals

TEZAN SAHU

services.tezansahu@gmail.com

A LITTLE ABOUT ME…

Hi, I'm Tezan.



Microsoft IDC (2021 - Present)

Sr. Applied Scientist · M365 Copilot Extensibility

10* US Patents · Multiple Research Papers

Author

"Beyond Code" — **National Bestseller**

"Low-Pass Filter" — AI Newsletter

IIT Bombay (2017 - 21)

Mech. Engg. (9.90 CPI) · **Institute Rank 2**

Minor in Computer Science & Engineering

Speaker & Mentor

TEDx & PyConf Speaker · **50+** Talks, Workshops & Panels

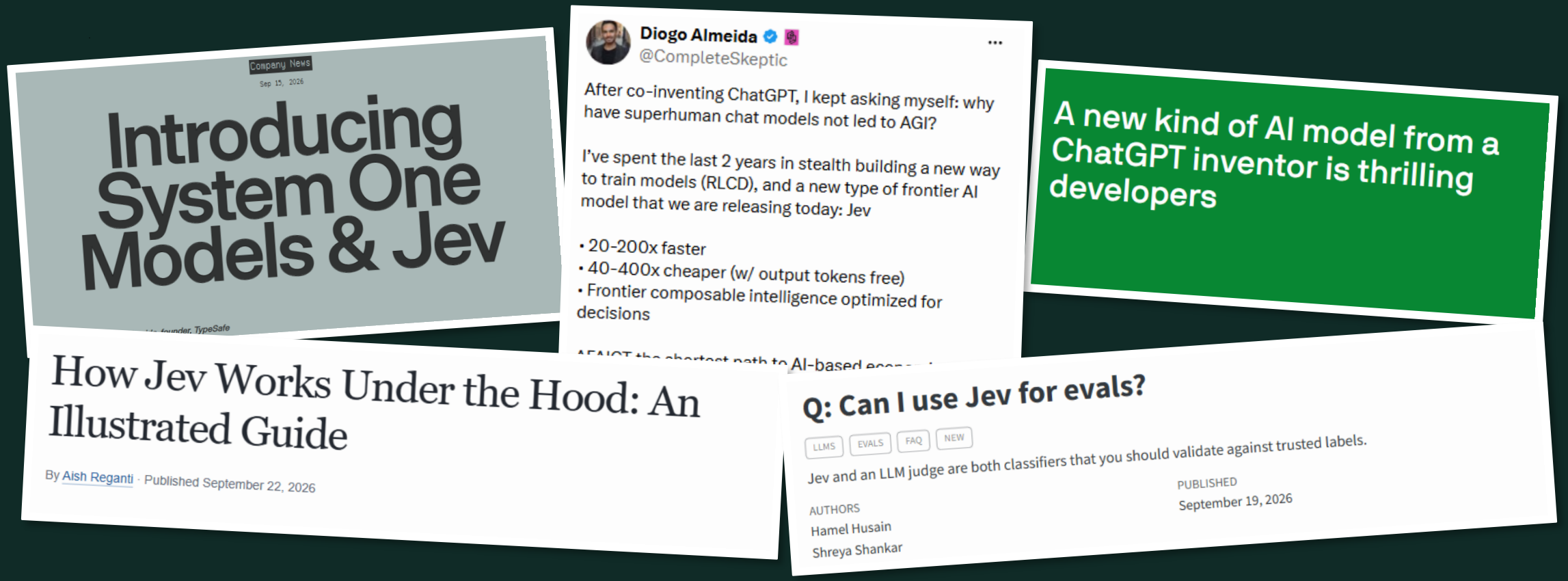
25,000+ students & professionals

OFF THE CLOCK…

- Trained vocalist
- Racket-sport nerd
- Artist · 3 exhibitions

services.tezansahu@gmail.com

Who's heard of Jev?



services.tezansahu@gmail.com

Meet Jev: a verdict, not another paragraph.

OUR WORKSHOP TRACE

"Everyone was notified."

event: draft_created

Does the evidence support it?

→ **Jev**
TypeSafe AI →

ALLOWED ANSWERS

supported

contradicted

insufficient evidence

+ probabilities

Code checks exact receipts. Jev can assess what the language claims from the evidence.

services.tezansahu@gmail.com

An impression, or a problem to solve?

“Wonderful. Third room change today.”

Pleased? Irritated? Not enough context?

READ THE TONE

A quick interpretation

REPLAN THE SESSION

Capacity + projector + availability

Thinking, Fast and Slow / Daniel Kahneman

Kahneman’s familiar framing / original chat illustration, not a brain-model equivalence

services.tezansahu@gmail.com

Same reader. Different answer sheet.

GENERATE WORDS

contextual representation

↓

vocabulary head

“I” → “have” → “sent” → …

Choose next token. Append. Repeat.

SCORE ALLOWED ANSWERS

contextual representation

↓

candidate scores → softmax

supported 0.12
insufficient 0.83
contradicted 0.05

Less output generation can save work. Reading the input is still work.

Conceptual mechanism; not an assertion about Jev’s private architecture

services.tezansahu@gmail.com

What can the application ask?

ONE INCIDENT. THREE DIFFERENT QUESTIONS.

STATE

assistant claim

tool events

available evidence

→

NOUL — P(yes): claimed completion?

CHOICE — supported / contradicted / insufficient

SCORE — ordered rubric + distribution

The application owns validation, thresholds, permissions and side effects.

services.tezansahu@gmail.com

One record. Several useful judgments.

GENERATIVE OUTPUT

{ “status”: “sent”, … }

token → **token** → **token**

A sequence grows one token at a time.

TYPED QUESTION OUTPUTS

Claimed completion? → P(yes)

Evidence status? → Choice

Overclaim severity? → Score

Independent questions return together.

A predictable response shape is useful. Meaningful probabilities require more.

Vendor-described API behavior; internal pass structure undisclosed

services.tezansahu@gmail.com

What behaviour does the training reward?

RLHF

“Which reply would you prefer?”

Human comparisons → learned reward → optimize responses.

RLVR

“Did the answer pass the check?”

Verifiable outcome → reward for solving the task.

RLCD / STATED GOAL

“Does 80% behave like 80%?”

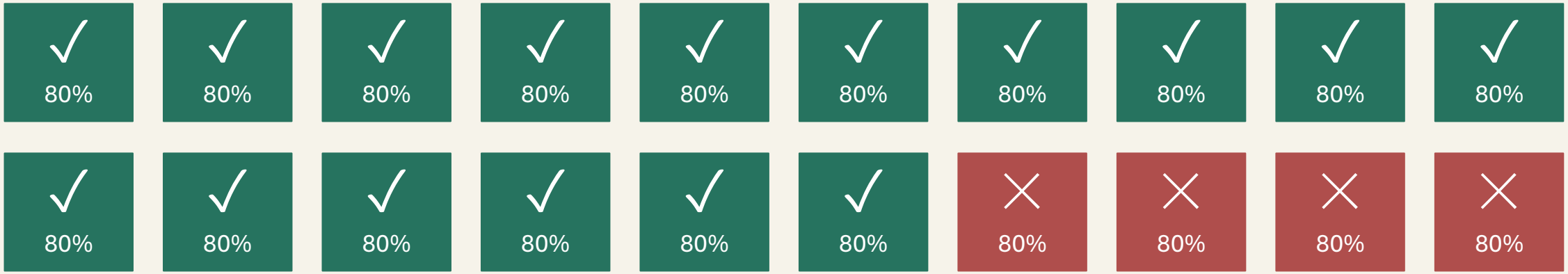
Learn useful bounded decisions with reliable probabilities.

Jev’s actual RLCD reward, optimizer, training recipe and architecture: unpublished.

Founder’s motivation; RLHF / RLVR / RLCD are contrasted objectives

services.tezansahu@gmail.com

What should “80%” actually mean?



16 correct

4 incorrect

If calibrated: about 16 in 20 across similar cases. Not a promise about one answer.

Illustrative outcomes, not a measured Jev calibration test

services.tezansahu@gmail.com

Make unjustified certainty expensive.

$$\text{reward} = \text{correctness} - (\text{confidence} - \text{correctness})^2$$

WRONG / 90%

−0.81

Large penalty

WRONG / 20%

−0.04

Smaller penalty

CORRECT / 90%

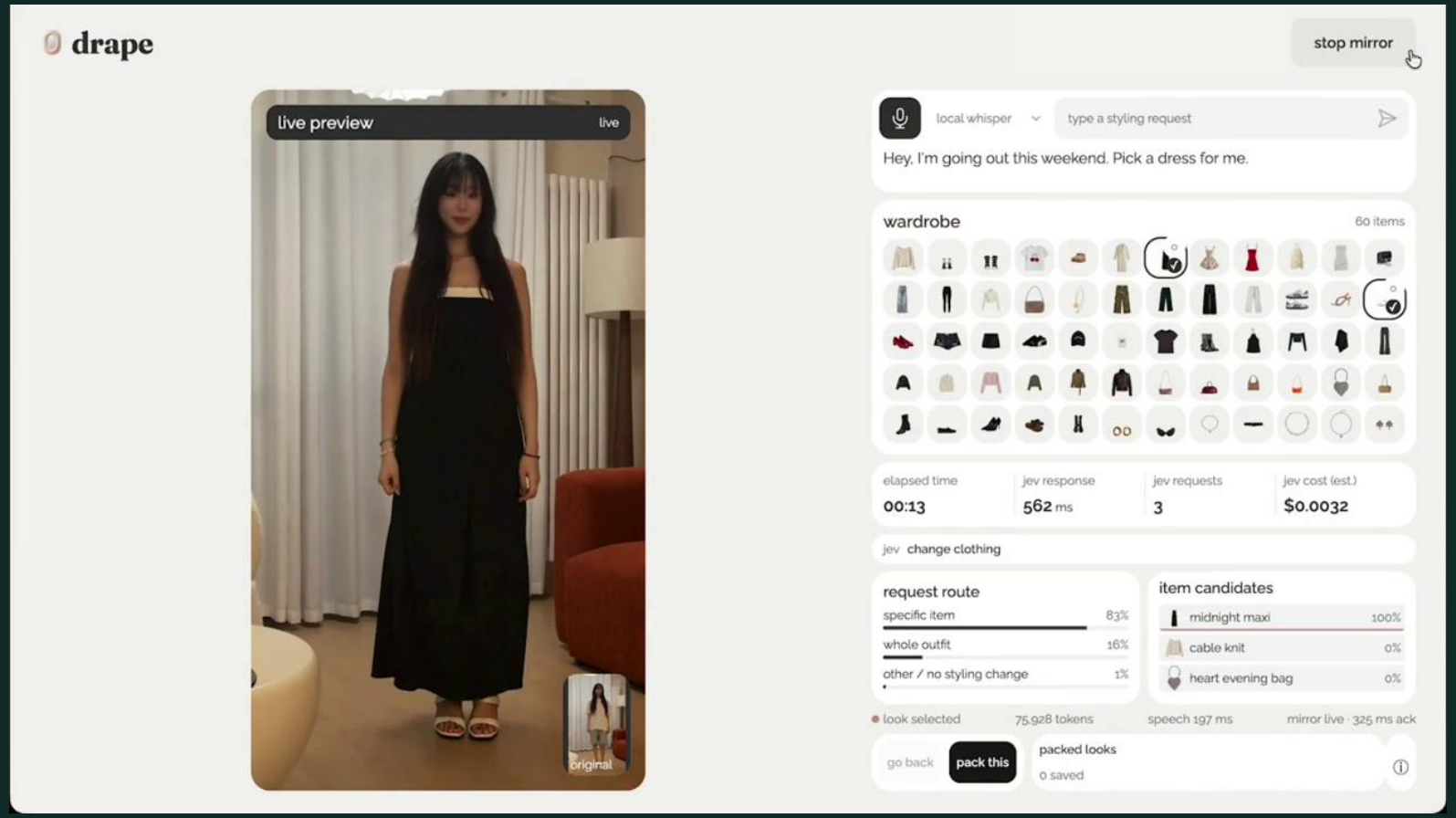
+0.99

High reward

RLCR illustrates one approach. No established identity with TypeSafe's RLCD.

RLCR: Beyond Binary Rewards / source-linked mathematical analogue

Choose an outfit from a spoken request.



Speech + closet context

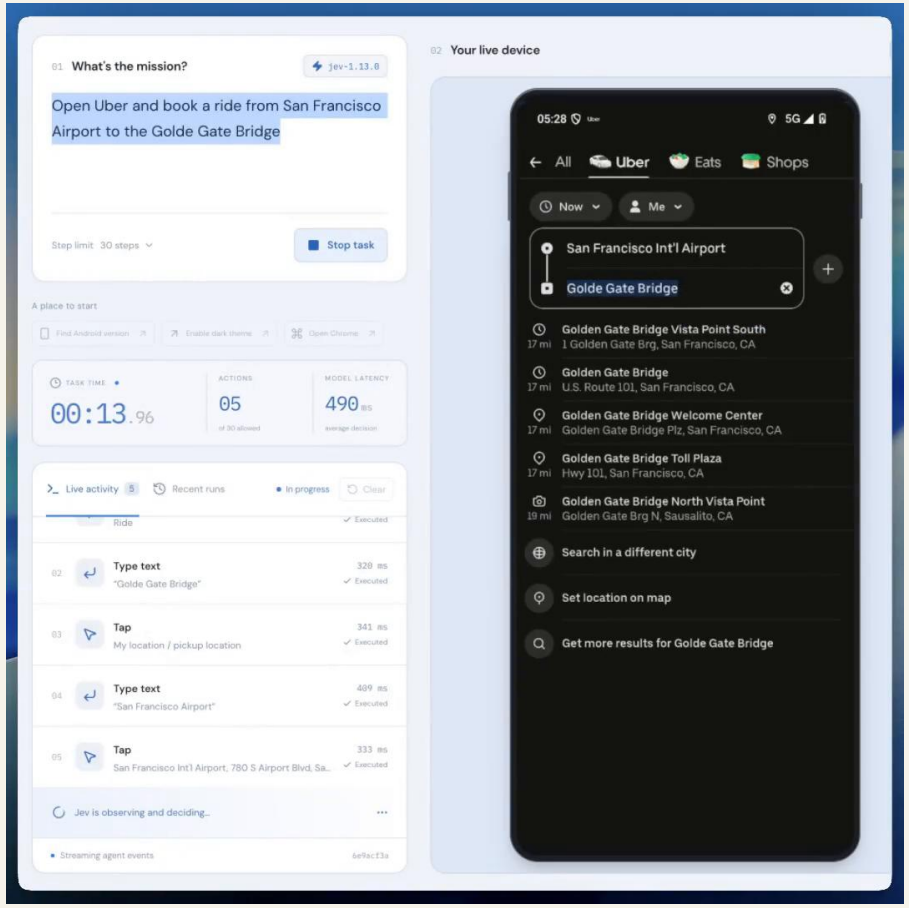
Jev selects the outfit.

A separate pipeline renders the try-on.

Original video: Nailthy Tang / @nailthy62 — Drape experiment

services.tezansahu@gmail.com

Choose the next step in a ride request.



TASK / SFO → GOLDEN GATE BRIDGE

Jev chooses
an action + an observed control.

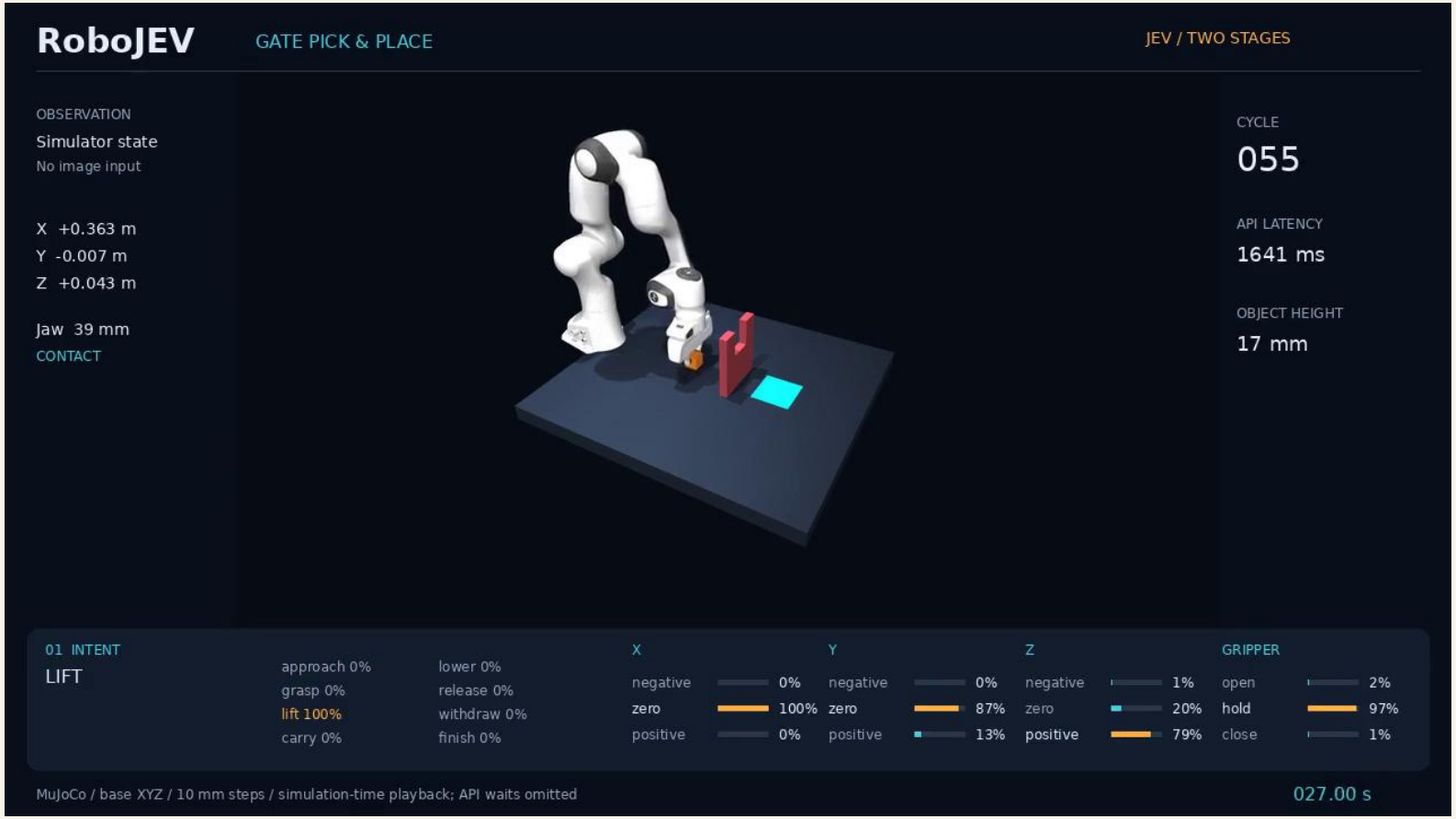
The app executes it,
then observes the next screen.

Shown: payment selection, not a completed booking.

Video: droidrun/mobile-jev (MIT) / direct source & full recording

services.tezansahu@gmail.com

Carry the object past the obstacle.



Jev chooses intent + motion.

A controller executes it.

Watch the clearance.

Failure excerpt.
Structured state, not vision.
API waits omitted.

Video: lykycy123/RoboJEV (Apache-2.0) / full experiment, code and recordings

services.tezansahu@gmail.com

The reusable part is the judgment.

BROWSER AGENT

Pick the next action

Jev chooses an observed control.
A small LLM writes field text.
Code validates and executes.

DATABASE PREDICATE

Ask a question of rows

“Which meeting notes contain
a commitment with no owner?”
Illustrative pg-jev question.

pg-jev / original project

A component can enable a rich experience without doing every part of it.

services.tezansahu@gmail.com

Was this happening last month too?

STORED HISTORY

- **conversations**
- **agent traces**

instrumented + retained

→

A NEW QUESTION

“Did the agent claim completion without supporting evidence?”

Question → judgment per eligible row

Cheap judgment expands the investigation. It does not validate the judge.

services.tezansahu@gmail.com

Which of these should we flag?

FINAL RESPONSE	AVAILABLE RECORD	HUMAN JUDGMENT
“Everyone was notified.”	Complete trace: draft only	**Flag the overclaim**
“Draft ready. Shall I send?”	Complete trace: draft only	**Do not flag**
“Everyone was notified.”	Relevant trace unavailable	**Need more evidence**

Rows 1–2 assume reliable send-event logging. Row 3 has no such assurance.

Define the failure before counting it. Missing evidence is not a clean bill of health.

services.tezansahu@gmail.com

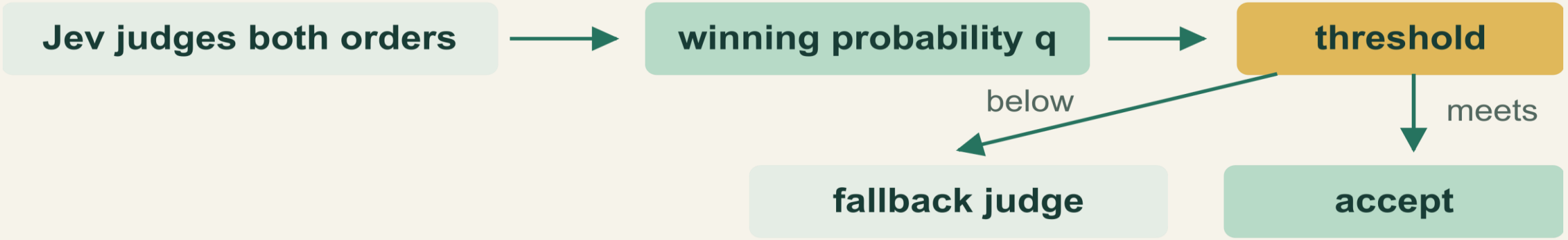
How much judgment can we delegate?

		JEV	COMPARATOR
Preference RewardBench / n=400	Truthful status beats a polished overclaim	92.2%	93.5%
Hard correctness JudgeBench / n=350	A code answer misses the empty-list case	78.6%	93.1%
No evidence HaluEval general / n=80	Certify a claim without its source	52.5%	55.0%

Examples are illustrative, not paper items. Comparator: GPT-6; last row GPT-5.4.

services.tezansahu@gmail.com

Let uncertainty trigger a second opinion.



q = max(pA, pB), after aligning and averaging both orders.

53.7%
accepted by Jev

92.5%
cascade accuracy

56.8%
reported fee ratio

GPT-6 alone: 93.1% accuracy. Another fallback's frozen threshold missed tolerance.

Frozen two-order policy / 510 held-out extension pairs / offline simulated cascade

services.tezansahu@gmail.com

But what if the wrong answer looks certain?

0.518

≈ chance at separating correct from incorrect

Uncertain? Ask again.
Confidently wrong?
The gate may miss it.

Valid shape ≠ true judgment ≠ authorized action.

Reference-free prose: confidence error-detection AUROC 0.518

services.tezansahu@gmail.com

Don’t wait for another empty room.

INSTANT EVALS	**Search history**	Find earlier unsupported completion claims
REGRESSION TEST	**Keep the cases**	Check the next version before releasing
ONLINE MONITOR	**Watch new runs**	Flag new overclaims after the response
BLOCKING GUARD	**Before replying**	Check receipts before claiming completion

LangWatch evaluation docs — online monitors run after the response, not before it

services.tezansahu@gmail.com

Which route could you use tomorrow?

TYPESAFE / HOSTED

Direct Jev

TypeSafe account + key
Published usage pricing
$0.042 / 1M input tokens

TypeSafe docs

BEATAPI / HOSTED

Free Jev route

jev-1.13-free
1 successful call/min
10/min after any top-up

BeatAPI terms

OLLAMA / LOCAL

Open alternatives

Nimble 9B
Tev1 4B / 0.8B
Ollama 0.35+

Ollama model list

Same wire contract ≠ same weights, limits, privacy or quality. No live access test performed.

Provider docs verified 4 Oct 2026 / access and terms can change

services.tezansahu@gmail.com

“Didn’t BERT already classify text?”

	TYPICAL BERT CLASSIFIER	JEV / TYPESAFE AI
Task	Train a head for a particular label set	Supply state, question + choices
New criterion	Usually adapt the task-specific classifier	Ask a new question; validate it
Probabilities	Class probabilities; calibrate separately	RLCD targets calibrated decisions

Zero-shot classifiers predate Jev. Neither probabilities nor runtime choices are new by themselves.

The proposition: a reusable, typed decision model — not a newly invented classification task.

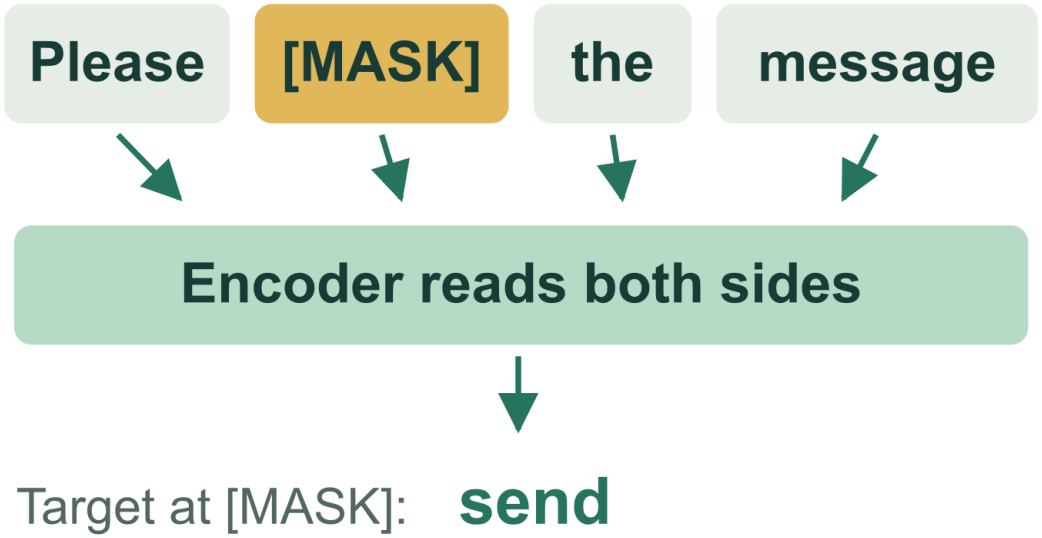
Reganti + Raschka / conventional task-specific BERT head versus Jev’s published interface

services.tezansahu@gmail.com

A decision model can use either backbone.

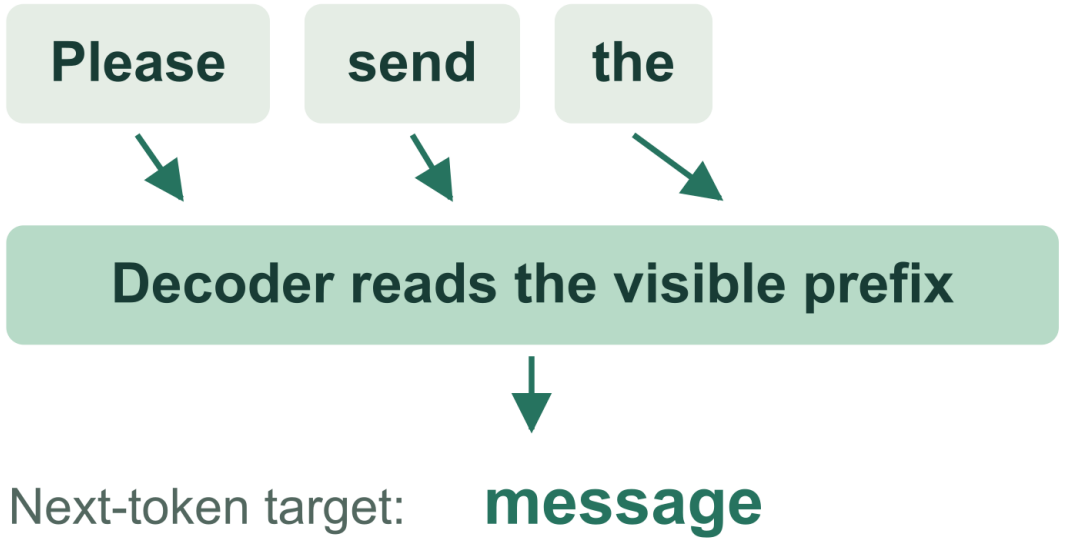
BERT-STYLE ENCODER

Fill a hidden token inside the input



CAUSAL DECODER

Predict the next token after the input



After pretraining: either family can support classification with a suitable head or scorer.

Hugging Face LLM course / pretraining examples, not Jev's disclosed architecture

services.tezansahu@gmail.com

Three open models. Different recipes.

OPEN-JEV

state + question
+ one candidate

↓

decoder + LoRA

↓

scalar decision head

LAYA

state + question
+ all marked options

↓

ModernBERT encoder

↓

shared option scorer

BESPOKE NIMBLE 9B

state + schema

↓

Qwen3.5-9B + LoRA

↓

score allowed answer tokens

Next: how Bespoke trains Nimble to notice the evidence that changes an answer.

Open-Jev + Laya: Reganti's guide / Bespoke Nimble: project README linked in notes

services.tezansahu@gmail.com

Nimble: learn what changes the answer.

CLAIM: "THE NOTIFICATION WAS SENT"

`draft_created`

NO SEND RECEIPT

Insufficient evidence

CLAIM: "THE NOTIFICATION WAS SENT"

`draft_created`

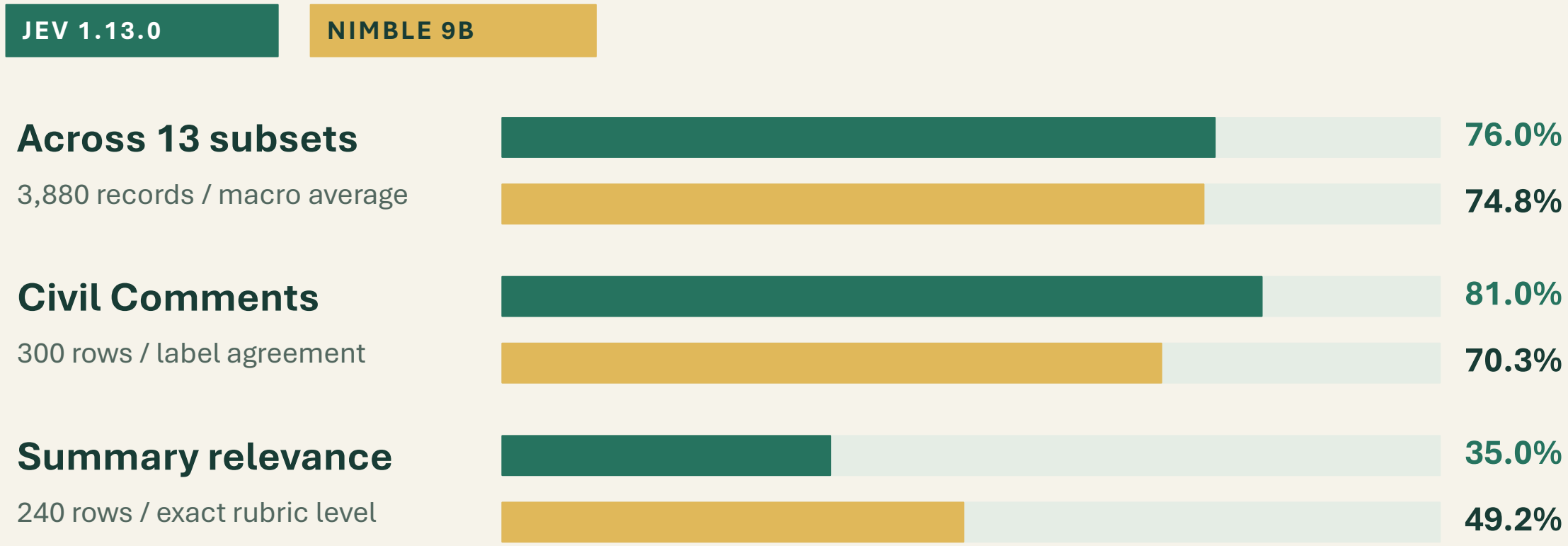
`+ send_success`

Evidence supports "sent"

Our illustrative pair: same wording, one decisive fact changed. Delivery is still separate.

services.tezansahu@gmail.com

Which model would you choose for this job?



Different tasks reverse the ranking. Author-run ≠ independent replication.

Bespoke's author-run public suite: identical rows / criteria / human reference labels

services.tezansahu@gmail.com

Try the same incident on a local model.

1 / DOWNLOAD

```
ollama pull nimble
```

2 / SEND TO LOCAL API

```
POST localhost:11434
/v1/systemone
```

STATE

```
claim: “notification sent”
events: [draft_created]
```

CHOICE / evidence status

```
supported
contradicted
insufficient
```

Conceptual request shape. Keep keys private; validate outputs; test your own cases.

Ollama decision endpoint / use HTTP or TypeSafe SDK, not ordinary chat mode

What does *“done”* mean in your own agent?

“MESSAGE SENT”

Find the receipt

A draft is not a send.
A send is not delivery.

“DOCUMENT UPDATED”

Check the change

Which file? Which version?
What was actually saved?

“TASK COMPLETED”

Define the finish

Which required steps?
Which outcomes remain unverified?

Label ~25 starting cases. Measure misses and false alarms. Keep a separate holdout.

services.tezansahu@gmail.com

Open models: compare the right things.

Open-Jev	Decoder + LoRA + scalar head	179 / 231; author JevBench subset
Laya	Encoder + option representations	Different tests; do not pool rankings
Nimble 9B	Contrastive fine-tuning / option logits	74.8% macro; shared 3,880-row suite
Tev1 4B / 0.8B	Ollama experimental decision models	Recipe / comparable score not asserted
Dohnuts	0.8B text + image fine-tune	65.8% of 231; non-commercial weights

See notes and research ledger for per-project evidence and licenses

services.tezansahu@gmail.com

A reading path, not a wall of links.

01 / MECHANICS	Illustrated guide + classifier history	1	2
02 / TRAINING TARGETS	TypeSafe + AI Engineer founder talk	1	2
03 / EVALS	Classifier validation + JEV-as-a-Judge	1	2
04 / STARTING POINTS	Instant Evals + Ollama decision models	1	2

services.tezansahu@gmail.com

TAKE THE QUESTION WITH YOU

Don’t ask if it sounds done.

Ask what would prove it.

That’s all, folks.

Questions welcome.



LinkedIn



Low-Pass Filter

services.tezansahu@gmail.com